

## Lists - Ordered List, Unordered List and Definition List

List-related tags are meant for marking up a list of items. HTML supports three types of lists: ordered list, unordered list and definition list.

### i. Unordered List `<ul>...</ul>` and List Item `<li>...</li>`

Function: An unordered list is shown with a bullet in front of each item. The `<ul>...</ul>` contains an unordered list. Each of items in the list is enclosed in `<li>...</li>`, as follows:

```
<ul>
  <li>list-item-1</li>
  <li>list-item-2</li>
  .....
</ul>
```

#### Example:

```
<p>An OMO web designer must master:</p>
<ul>
  <li>Hypertext Markup Language (HTML)</li>
  <li>Cascading Style Sheet (CSS)</li>
  <li>HyperText Transfer Protocol (HTTP)</li>
  <li>Apache HTTP Server</li>
</ul>
```

**Output of the example:** An OMO web designer must master:

1. Hypertext Markup Language (HTML)
2. Cascading Style Sheet (CSS)
3. HyperText Transfer Protocol (HTTP)
4. Apache HTTP Server

### ii. Ordered List `<ol>...</ol>` and List Item `<li>...</li>`

Items in an ordered list are numbered automatically by the browser. The container tag `<ol>...</ol>` contains an ordered list. Each item of the list is contained inside a `<li>...</li>` container tag. The syntax is similar to the unordered list.

#### Example:

```

<p>An OMO web designer must master:</p>
<ol>
  <li>Hypertext Markup Language (HTML)</li>
  <li>Cascading Style Sheet (CSS)</li>
  <li>HyperText Transfer Protocol (HTTP)</li>
  <li>Apache HTTP Server</li>
</ol>

```

### iii. Definition List <dl>...</dl>, Definition Term <dt>...</dt> and Definition Detail <dd>...</dd>

Function: <dl>...</dl> tag contains a Definition List. Each of <dt>...</dt> and <dd>...</dd> pair contains a Definition Term and the Definition Detail, as follows:

```

<dl>
  <dt>term-1</dt>
  <dd>definition-for-term-1</dd>
  <dt>term-2</dt>
  <dd>definition-for-term-2</dd>
  .....
</dl>

```

#### Example:

```

<p>These are some of the commonly-encountered web protocols:</p>
<dl>
  <dt>HTTP</dt><dd>HyperText Transfer Protocol</dd>
  <dt>FTP</dt><dd>File Transfer Protocol</dd>
  <dt>SMTP</dt><dd>Simple Mail Transfer Protocol</dd>
  <dt>POP</dt><dd>Post Office Protocol</dd>
  <dt>PPP</dt><dd>Point-to-point Protocol</dd>
  <dt>SLIP</dt><dd>Serial Line Internet Protocol</dd>
</dl>

```

**Output of the example:** These are some of the commonly-encountered web protocols:

**HTTP:** HyperText Transfer Protocol

**FTP:** File Transfer Protocol

**SMTP:** Simple Mail Transfer Protocol

**POP:** Post Office Protocol

**PPP:** Point-to-point Protocol

**SLIP:** Serial Line Internet Protocol

The "unordered list" and "ordered list" are used in most of the HTML documents. But I don't find many web pages using the "definition list".

#### **iv. Nested Lists**

##### **Example 1:**

```
<p>The topics covered are:</p>
<ul>
  <li>HyperText Markup Language (HTML)
    <ul>
      <li>Based on SGML</li>
      <li>Used to create web pages</li>
      <li>Maintained by W3C</li>
    </ul>
  </li>
  <li>Cascading Style Sheet (CSS)
    <ul>
      <li>Used to define presentation style for web pages</li>
      <li>Also maintained by W3C</li>
    </ul>
  </li>
</ul>
```

##### **Example 2:**

```
<ol>
  <li>XML: Extensible Markup Language
    <ol>
      <li>Based on SGML</li>
      <li>Maintained by W3C</li>
    </ol>
  </li>
  <li>DOM: Document Object Model</li>
  <li>SAX: Simple API for XML</li>
</ol>
```

## Tables

Table-related tags are meant for tabulating data. (Older HTML documents tend to use `<table>` for formatting the document to divide the document into columns/sections, which should be avoided. Use style sheets for formatting instead.)

The basic unit of a table is a cell. Cells are grouped into rows. Rows are grouped to form the table. This corresponds well to the "row-centric" approach in the display.

The tags used by table are:

- `<table>...</table>`: contains the entire table.
- `<tr>...</tr>`: contains a row.
- `<th>...</th>` and `<td>...</td>`: contain a header cell and a data (detail) cell respectively.
- `<caption>...</caption>`: specifies a caption.
- `<thead>...</thead>`, `<tbody>...</tbody>`, and `<tfoot>...</tfoot>`: for table header, body, and footer.
- `<colgroup>...</colgroup>` and `<col>...</col>`: for applying styles to column group and column respectively.

**For Example:**

```
<table>
  <caption>Price List</caption>
  <tr>
    <th>Fruit</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Apple</td>
    <td>$0.50</td>
  </tr>
  <tr>
    <td>Orange</td>
    <td>$0.65</td>
  </tr>
</table>
```

## Table `<table>...</table>`

Function: Set up a table, consisting of rows of cells. Three optional presentation attributes, `border="n"`, `cellspacing="n"`, and `cellpadding="n"`, are often used in older HTML pages but now deprecated. The now-preferred approach is to use CSS.

## Table Row `<tr>...</tr>`

Function: Set up a row inside a table, consisting of cells.

## Table Header Cell `<th>...</th>`, Table Data Cell `<td>...</td>`

Function: Set up each individual cell of a row. `<th>...</th>` defines a header cell (usually displayed in bold with center alignment) and `<td>...</td>` defines a body cell.

## Attributes "rowspan" and "colspan"

On a `<td>` cell, we can use the attribute `rowspan="numOfRows"` or `colspan="numOfColumns"` to span the cell to occupy multiple rows or columns.

### Example 1:

```
<table>
  <tr>
    <td>11111</td>
    <td>22222</td>
    <td>33333</td>
  </tr>
  <tr>
    <td>44444</td>
    <td rowspan="2">55555</td>
    <td>66666</td>
  </tr>
  <tr>
    <td>77777</td>
    <td>88888</td>
  </tr>
</table>
```

### Example 2:

```

<table>
  <tr>
    <td colspan="2" rowspan="2">11111</td>
    <td>22222</td>
  </tr>
  <tr>
    <td>33333</td>
  </tr>
  <tr>
    <td>44444</td>
    <td>55555</td>
    <td>66666</td>
  </tr>
</table>

```

## Table Header <thead>...</thead>, Table Body <tbody>...</tbody> and Table Footer <tfoot>...</tfoot>

Function: Used to define a header, body and footer sections for a table. Browsers may duplicate the header or footer when breaking the table across multiple pages.

## Column Group <colgroup>...</colgroup> and Column <col />

Function: <colgroup>...</colgroup> can be used to group a set of columns so that style can be applied to all. The attribute `span="numOfColumns"` specifies the number of columns.

### Example 1:

```

<table>
  <!-- col group 1 spans 3 columns -->
  <colgroup span="3" style="background-color:lightgrey"></colgroup>
  <tr>
    <td>Col 1 is in the group</td>
    <td>Col 2 is in the group</td>
    <td>Col 3 is in the group</td>
    <td>Col 4 is NOT in the group</td>
  </tr>
  <tr>
    <td>Col 1 is in the group</td>

```

```

        <td>Col 2 is in the group</td>
        <td>Col 3 is in the group</td>
        <td>Col 4 is NOT in the group</td>
    </tr>
</table>

```

### Example 2:

```

<table>
  <!-- colgroup 1 consists of col 1 and col 2 -->
  <colgroup>
    <col style="background-color:lightyellow" />
    <col style="background-color:white" />
  </colgroup>
  <!-- colgroup 2 consists of col 3 and col 4 -->
  <colgroup style="background-color:lightgrey" >
    <col span="2" />
  </colgroup>
  <tr>
    <td>Col 1 in the group 1</td>
    <td>Col 2 in the group 1</td>
    <td>Col 3 in the group 2</td>
    <td>Col 4 in the group 2</td>
  </tr>
  <tr>
    <td>Col 1 in the group 1</td>
    <td>Col 2 in the group 1</td>
    <td>Col 3 in the group 2</td>
    <td>Col 4 in the group 2</td>
  </tr>
</table>

```

## Anchor and Link <a>...</a>

A hypertext-link allows users to:

1. Navigate to a different document.
2. Navigate to an "Anchor Point" (bookmark).
3. Request other Internet resources (such as e-mail).

The anchor tag <a>...</a> can perform one of these two functions:

1. It can be used to set up a hyperlink.
2. It can be used to set up a "named anchor point" (bookmark).

## Videos and Audio Files

The easiest way to add video or sound is to include the `<embed>` tag. You can also include a `<noembed>` tag for older browsers.

### Attributes:

- **Align:** Determines object alignment (center, left, right).
- **Autostart:** Boolean to start automatically.
- **Loop:** Specifies continuous play.
- **Playcount:** Number of times to play (IE alternative for loop).
- **Hidden:** Hide or show the object.
- **Src:** URL of the object.
- **Volume:** Controls volume (0-100).

### Embed Video Example:

```
<video controls width="250">  
  <source src="/media/cc0-videos/flower.mp4"  
    type="video/mp4">  
</video>
```

### Embed Audio Example:

```
<audio controls src="/media/cc0-audio/t-rex-roar.mp3">  
</audio>
```

## HTML Forms

HTML forms are required to collect data from visitors (e.g., login, registration).



## Form Element

Wraps all other elements.

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign Up Form</title>
    <link rel="stylesheet" href="css/normalize.css">
    <link href="https://fonts.googleapis.com/css?family=Lato:300,400,700" rel="stylesheet">
    <link rel="stylesheet" href="css/main.css">
  </head>
  <body>
    <form action="index.html" method="post">

    </form>
  </body>
</html>
```

### Attributes:

1. **action:** The URL of the program that processes the information.
2. **method:** The HTTP method (POST or GET).

## Input Element

Used to make text fields.

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign Up Form</title>
    <link rel="stylesheet" href="css/normalize.css">
    <link href="https://fonts.googleapis.com/css?family=Lato:300,400,700" rel="stylesheet">
    <link rel="stylesheet" href="css/main.css">
  </head>
  <body>
    <form action="index.html" method="post">
```

```

        <input type="text" id="name" name="student_name">
    </form>
</body>
</html>

```

#### Attributes:

- **type:** Kind of input (text, email, password, etc.).
- **id:** Used to associate labels or for CSS/JS.
- **name:** Necessary for server-side processing.
- **placeholder:** Short hint displayed in the field.

## Textarea Element

Used for multiple lines of text.

```

<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign Up Form</title>
    <link rel="stylesheet" href="css/normalize.css">
    <link href="https://fonts.googleapis.com/css?family=Lato:300,400,700" rel="stylesheet">
    <link rel="stylesheet" href="css/main.css">
  </head>
  <body>
    <form action="index.html" method="post">
      <textarea id="bio" name="student_bio"></textarea>
    </form>
  </body>
</html>

```

## Button Element

Used to submit forms. Types include submit, reset, and button.

```

<!DOCTYPE html>
<html>
  <head>

```

```

    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign Up Form</title>
    <link rel="stylesheet" href="css/normalize.css">
    <link href="https://fonts.googleapis.com/css?family=Lato:300,400,700" re
    <link rel="stylesheet" href="css/main.css">
</head>
<body>
    <form action="index.html" method="post">

        <input type="text" id="name" name="student_name">
        <input type="email" id="mail" name="student_email">
        <input type="password" id="password" name="student_password">

        <textarea id="bio" name="student_bio"></textarea>

        <button type="submit">Submit</button>

    </form>
</body>
</html>

```

## Label Element

Identifies form controls. The `for` attribute matches the `id` of the input.

```

<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign Up Form</title>
    <link rel="stylesheet" href="css/normalize.css">
    <link href="https://fonts.googleapis.com/css?family=Lato:300,400,700" re
    <link rel="stylesheet" href="css/main.css">
  </head>
  <body>
    <form action="index.html" method="post">
      <label for="name">Name</label>
      <input type="text" id="name" name="student_name">

```

```

        <label for="mail">Email</label>
        <input type="email" id="mail" name="student_email">

        <label for="password">Password</label>
        <input type="password" id="password" name="student_password">

        <label for="bio">Biography</label>
        <textarea id="bio" name="student_bio"></textarea>

        <button type="submit">Submit</button>
    </form>
</body>
</html>

```

## Select Menus

Dropdown list of options.

```

<!DOCTYPE html>
<html>
    <head>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Sign Up Form</title>
        <link rel="stylesheet" href="css/normalize.css">
        <link href="https://fonts.googleapis.com/css?family=Lato:300,400,700" rel="stylesheet">
        <link rel="stylesheet" href="css/main.css">
    </head>
    <body>
        <form action="index.html" method="post">
            <h2>Your basic info</h2>

            <label for="name">Name</label>
            <input type="text" id="name" name="student_name">

            <label for="mail">Email</label>
            <input type="email" id="mail" name="student_email">

            <label for="password">Password</label>
            <input type="password" id="password" name="student_password">

```

```

<h2>Your profile</h2>

<label for="bio">Biography</label>
<textarea id="bio" name="student_bio"></textarea>

<label for="courses">Select Courses</label>
<select id="courses" name="student_courses">
  <optgroup label="Engineering">
    <option value="computer_engineering">Computer Science</option>
    <option value="electrical_engineering">Electrical</option>
    <option value="mechanical_engineering">Mechanical</option>
    <option value="chemical_engineering">Chemical</option>
    <option value="civil_engineering">Civil</option>
  </optgroup>
  <optgroup label="Management">
    <option value="finance_management">Finance Management</option>
    <option value="technology_management">Technology</option>
    <option value="marketing_management">Marketing</option>
    <option value="business_administration">Business</option>
  </optgroup>
</select>

  <button type="submit">Submit</button>
</form>
</body>
</html>

```

## Radio Buttons

Used for picking one option from a small group.

```
<input type="radio" id="under_16" value="under_16" name="user_age">
```

## Checkboxes

Used for selecting multiple options.

```
<input type="checkbox" id="engineering" value="interest_engineering" name="user_
```

## **Difference between Checkbox and Radio button**

1. Checkbox can exist on its own, while radio buttons can only appear as a group (minimum of 2 radio buttons should be there).
2. Selecting checkbox is optional but choosing one of the radio button is mandatory.